



ME2 – Voice Controlled Smart Device



Rowel Atienza • Sep 12

100 points | Due Sep 26, 1:59 PM

ASR models are not desirable for on-device computing because of footprint. The goal of this ME is to build a tiny Voice Command Model (VCM) that can understand the most common commands humans tell their smart devices:

Most common commands issued to smart devices, enumerated

Ranked by real-world log and survey data (259,164 logged commands from Amazon Alexa + Google Home devices, plus recurring U.S. consumer surveys):

Top commands, by rank

1. ****Play music**** — the #1 use case in every survey year (2018–2020). Actual forms: "play music"
2. ****Ask a question / search**** — **"what's the weather"** (daily-use #2–3), **"what time is it"**,
3. ****Control lights (IoT)**** — 85% of all Alexa IoT commands: **"turn on/off lights"**,
4. ****Dim / color lights**** — ~10% of IoT commands: **"Dim lights to X percent"**
5. ****Set a timer**** — **"set a timer for X minutes"**
6. ****Set an alarm**** — **"set an alarm for X am/pm"**
7. ****Adjust thermostat temperature**** — **"set temperature to X degrees"**
8. ****Media control**** — **"pause"**, **"stop"**, **"next/skip"**, **"volume up/down"**, **"louder"**. No. 1 can be folded to No. 8.
9. ****Reminders and lists**** — **"remind me to..."** "what are my reminders"
10. ****Calls and messaging**** — **"call mom"**

Your tasks:

1. Build a dataset to train VCMs - collective work
2. Build and train a VCM on this dataset - individual work
3. Design a benchmark for validating VCMs - collective work
4. Validate the performance of your VCM - individual work
5. Build a real-world demo of your VCM (this can run on RPi4/5 device) - individual work (may share devices)
6. Your VCM must be tiny - can run on RPi4 or 5 in real-time - individual work
7. VCM should be standalone - can not call cloud-based models.
8. No LLM, just pure VCM doing 1 to 10. Everything on-device.